

KITEWRIGHT RANGE · PRECIS

Kitewright Range design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3 numbers from KWR-CAL-001 v0.2 on the constructable design (KWR-DDR-002)
0.2	2026-10-03	Amish Chadha	TRL 2 precis with first-order numbers, components, design choices and safety (KWR-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

Kitewright Range design precis

An all-electric quadplane frame for the Kitewright family that takes off vertically and cruises on a fixed wing for long surveys.

■ Summary

Kitewright Range is a 2.5 m span quadplane of 10.6 kg that lifts off on four fixed 20 in rotors, flies its survey on the wing and lands vertically again. On paper it hovers at 5,000 m with a 31.8 % thrust margin and flies 31 min and 40.9 km on the wing with a 1 kg payload and a 20 % reserve. It is built from hot-wire cut foam and glass, lite-ply, carbon tubes and printed parts with hobby tools, and costs an estimated USD 3,945 against a USD 5,000 value-engineering target. Endurance and take-off mass miss their targets and the hover margin and range are thin; these are open decisions for Amish (docs/06-design-decisions.md).

■ How it works

A two-piece wing on a carbon joiner sits at the shoulder of a lite-ply fuselage. Under each wing a printed pylon holds a carbon boom with one lift motor 506 mm ahead of the centre of gravity and one 506 mm behind it, so the four rotors lift through the balance point; their discs run 81 mm below the wing-top plane. A tractor motor with a folding propeller at the nose drives the aircraft in cruise, and a conventional tail on a carbon boom steers it. The aircraft rises on the rotors, accelerates on the cruise motor to about 22 m/s, stops the rotors with their blades parked fore and aft, and flies a planned survey; then it slows, hovers and lands on four faired legs. Two ColdCell packs sit either side of the wing joiner on the Kitewright Core power bus, the Core avionics sit in the front bay and the payload hangs on the Core mount under the centre of gravity. The flight code is stock PX4 quadplane firmware with published parameters (KWR-DDR-001, D1).

■ Main components

Table 1. Components (numbers are BOM lines)

#	Component	Made from, or bought	Role
1, 2	Fuselage box and hatch	Lite-ply and birch ply, glass outside	Carries the wing, packs, avionics, payload mount and tail
3	Nose cone and cruise motor mount	Printed ASA	Fairs the nose and holds the cruise motor on the thrust line
4, 14, 15	Tail boom socket, tail boom, tail mount	Printed ASA, carbon tube	Carries the tail 1 m behind the wing
5, 6, 7, 8	Wing panels with ailerons, spars, joiner, wing bolts	XPS foam and glass, carbon tubes, nylon bolts	Lift in cruise; panels come off for transport
9, 10, 11, 12	Boom pylons, pylon bolts, clamp caps, lift booms	Printed ASA, steel bolts, carbon tube	Hold the lift motors below the wing
13	Landing legs	Carbon rod, printed clamp and fairing, rubber foot	Stand the aircraft on rough ground
16, 17	Stabiliser with elevator, fin with rudder	XPS foam and glass	Pitch and yaw stability and control
18 to 21	Motor mounts, ESCs, lift motors, 20 in propellers	Bought	Vertical take-off, hover and landing
22, 23	Cruise motor and ESC, 14 in folding propeller	Bought	Wing-borne flight
24, 25	Servos, heated pitot and airspeed sensor	Bought	Control surfaces; airspeed in cold and icing
26, 27	Kitewright Core avionics and payload mount	From the Kitewright Core	Autopilot, GNSS, radios, payload interface
28	ColdCell packs, two	From the ColdCell design	583 Wh of heated Li-ion energy

■ Numbers from the TRL 3 calculations

Table 2. Key figures (KWR-CAL-001 v0.2)

Quantity	Value
Span, wing area, aspect ratio	2.50 m, 0.704 m ² , 8.9
Take-off mass with 1 kg payload	10.58 kg
Centre of gravity	28.4 % of the 281 mm mean chord
Hover power and thrust margin at 5,000 m	1,815 W; 31.8 %
Stall and cruise speed at 5,000 m	17.6 m/s; 21.9 m/s
Lift-to-drag ratio in cruise	7.5
Cruise power from the packs	585 W
Energy for cruise after hover, climb and reserve	302 Wh
Endurance and range on the wing	31.0 min; 40.9 km
Lowest structural safety factor	2.9 (wing spar at 2.5 g)
Longest part for transport	1,205 mm
Estimated cost	USD 3,945; value-engineering target USD 5,000 (USD 1,055 under)

Assumptions are stated in KWR-CAL-001, Table 1. The most important are the maker-class thrust of the lift motors (56.9 N each at sea level, scaled by density) and a zero-lift drag coefficient of 0.077.

■ Key design choices

All are decided under Amish's 2026-10-03 pre-approval and argued in KWR-DDR-001 and KWR-DDR-002:

- PX4 quadplane firmware, as the Kitewright Core (D1).
- Two-piece 2.5 m wing on a joiner, sized to the 1.3 m transport case (D2, C6).
- One set of 20 in lift propellers for every altitude band (D3).
- Tractor cruise motor at the nose with a folding propeller, conventional tail on a tail boom (D5, C1, C5).
- Lift booms on printed pylons 90 mm below the wing, motors centred on the centre of gravity (C2 to C4).
- Two ColdCell 6S3P Li-ion packs either side of the joiner (D6, C7).
- Foam, glass, lite-ply, carbon tube and printed parts: buildable with hobby tools (D7).
- Staged powered tests, tie-downs and conservative failsafes (D9, D10).

■ Patent design-arounds

From the preliminary patent, trademark and prior-art screen (not legal advice):

- All-electric, single fixed wing, fixed lift and cruise motors (no tilting thrust), to stay clear of US10343774B2 (tilting thrust, to about 2037).
- Lift booms on pylons below the wing with the rotor discs 81 mm below the wing-top plane, to stay clear of US9120560B1 (rotors even with the wing top surfaces, to 2033). The model checks this on every run.

- Layout kept distinct from US9242738B2 (Wisk, to about 2032); claims still to be read by an attorney before any sale.
- Not a tail-sitter, keeping clear of the Airbound tail-sitter noted in the family screen.

■ Relationship to other lab projects

- Kitewright Core: autopilot, radios, GNSS mast, payload mount and power bus, used unchanged. Envelopes and interfaces assumed here are listed in `docs/REVIEW.md`.
- ColdCell: the two heated packs. Kitewright Lift shares the 6S bus.
- AltiRig: measures the lift motor and propeller thrust at 5,000 m density before any flight.
- LakeWatch: the first payload, within 1 kg on the Core mount.

■ Safety

Safety: Kitewright Range is published as an open engineering reference, not certified aviation equipment, for civilian use only. Certification under FAA Part 107, EASA rules or India's Drone Rules 2021 is out of scope at this TRL and will be addressed if prototypes progress.

Propellers. Four 20 in lift propellers and a 14 in cruise propeller can cause serious injury. The lift propellers are fitted only at the build plan's propeller safety stop, never on the bench; a removable arming plug and a lockable arming switch keep the motors dead while anyone handles the aircraft; first hovers are tied down to ground anchors inside a 15 m keep-out zone (KWR-DDR-001, D9).

Lithium-ion packs. Two packs hold about 580 Wh. Charge only between 0 and 45 °C pack temperature, in a fire-resistant box, never unattended; treat any pack from a crash as damaged and isolate it outdoors.

Loss of link or power in remote terrain. Return to launch on the wing and land vertically; land in place at the reserve; fly inside a geofence (D10). Fly only where local rules allow, away from people, aircraft and wildlife; high-altitude sites may lie in restricted border or protected areas.

Structure. The wing spar has the smallest margin (2.9 at 2.5 g). Proof-load the wing before the first flight (TRL 4).

■ Open questions

None in this precis. Requirements not met or at risk are open decisions for Amish in the design decisions register (`docs/06-design-decisions.md`).